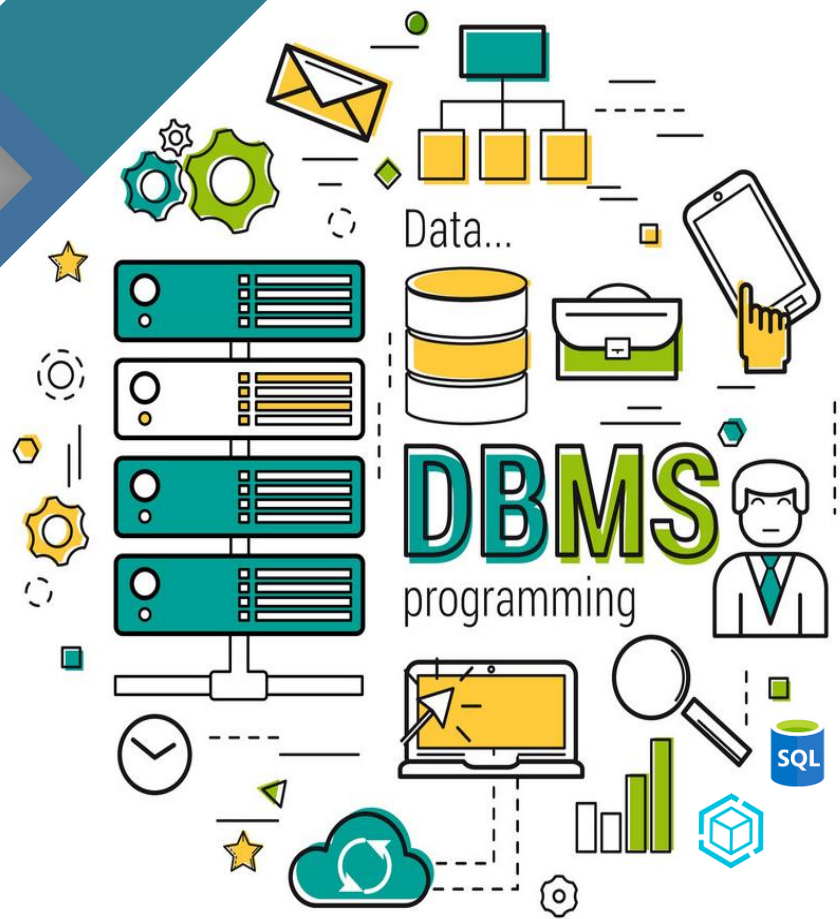




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Unit-10 Query Processing & Query Optimization

DATABASE MANAGEMENT SYSTEM



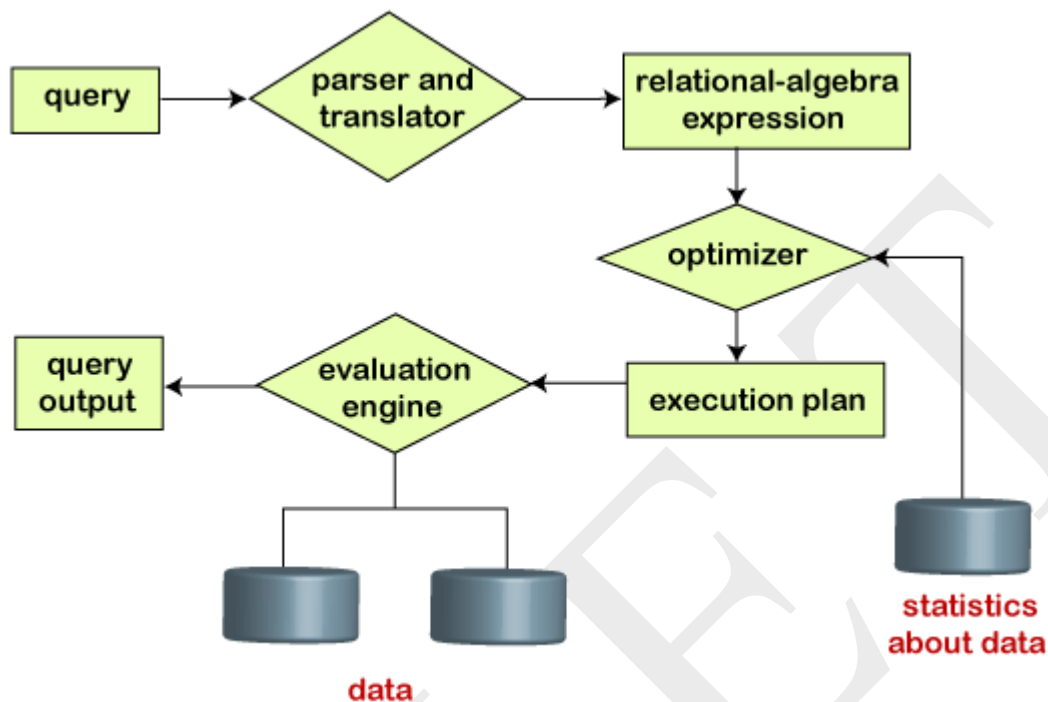
Unit 10	Query Processing & Query Optimization
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Query Processing Overview

- The steps involved in processing a query appear in Figure. The basic steps are:



1. Parsing and translation
2. Optimization
3. Evaluation

1. Parsing and Translation

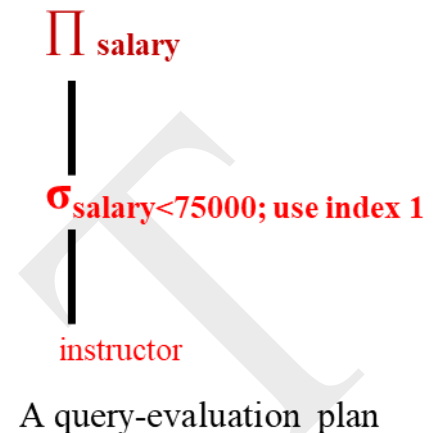
- The first action the system must take in query processing is to translate a given query into its internal form.
- In generating the internal form of the query, the parser checks the syntax of the user's query, verifies that the relation names appearing in the query are names of the relations in the database, and so on.
- The system constructs a parse-tree representation of the query, which it then translates into a relational-algebra expression.
- If the query was expressed in terms of a view, the translation phase also replaces all uses of the view by the relational-algebra expression that defines the view.

2,3. Optimization and Evaluation

- Given a query, there are generally a variety of methods for computing the answer. As an illustration consider the query.
Select salary **from** instructor **where** salary < 75000;
- This query can be translated into either of the following relational-algebra expression.

$$\sigma_{\text{salary} < 7500} \left(\prod \text{salary} (\text{instructor}) \right)$$
$$\prod \text{salary} \left(\sigma_{\text{salary} < 7500} (\text{instructor}) \right)$$

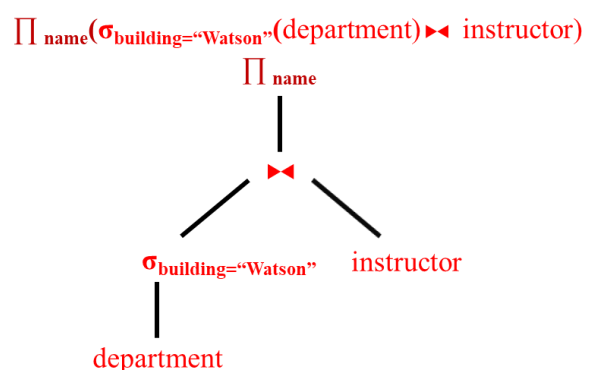
- Further, we can execute each relational-algebra operation by one of several different algorithms. For example, to implement the preceding selection, we can search every tuple in instructor to find tuples with salary less than 75000. If a B+-tree index is available on the attribute salary, we can use the index instead to locate the tuples.
- To specify fully how to evaluate a query, we need not only to provide the relational-algebra expression, but also to annotate it with instructions specifying how to evaluate each operation. **Annotations may state the algorithm to be used.**
- A sequence of primitive operations that can be used to evaluate a query is a query-execution plan or query-evaluation plan. The above figure illustrates an evaluation plan for our example query, in which a particular index (denoted in the figure as “index 1”) is specified for the selection operation.
- The query-execution engine takes a query-evaluation plan, executes that plan, and returns the answers to the query
- The different evaluation plans for a given query can have different costs. We do not expect users to write their queries in a way that suggests the most efficient evaluation plan. Rather, it is the responsibility of the system to construct a query evaluation plan that minimizes the cost of query evaluation; this task is called query optimization. Once the query plan is chosen, the query is evaluated with that plan, and the result of the query is output.



Evaluation of Expressions

Materialization

- It is easiest to understand intuitively how to evaluate an expression by looking at a pictorial representation of the expression in **an operator tree**. Consider the expression:
- If we apply the materialization approach, we start from the lowest-level operations in the expression (at the bottom of the tree).
- In our example, there is only one such operation: the selection operation on department. The inputs to the lowest-level operations are relations in the database.
- We execute these operations by the algorithms that we studied earlier, and we store the results in temporary relations.



- We can use these temporary relations to execute the operations at the next level up in the tree, where the inputs now are either temporary relations or relations stored in the database.
- In our example, the inputs to the join are the instructor relation and the temporary relation created by the selection on department. The join can now be evaluated, creating another temporary relation.
- By repeating the process, we will eventually evaluate the operation at the root of the tree, giving the final result of the expression. In our example, we get the final result by executing the projection operation at the root of the tree, using as input the temporary relation created by the join.
- Evaluation as just described is called **materialized evaluation**, since the results of each intermediate operation are created (materialized) and then are used for evaluation of the next-level operations.

Pipelining

- We can improve query-evaluation efficiency by reducing the number of temporary files that are produced. We achieve this reduction by combining several relational operations into a pipeline of operations, in which the results of one operation are passed along to the next operation in the pipeline. Evaluation as just described is called **pipelined evaluation**.
- **Creating a pipeline of operations can provide two benefits:**
 1. It eliminates the cost of reading and writing temporary relations, reducing the cost of query evaluation.
 2. It can start generating query results quickly, if the root operator of a query evaluation plan is combined in a pipeline with its inputs. This can be quite useful if the results are displayed to a user as they are generated, since otherwise there may be a long delay before the user sees any query results.
- **Pipelines can be executed in either of two ways:**
 1. In a **demand-driven pipeline**, the system makes repeated requests for tuples from the operation at the top of the pipeline.
 - Each time that an operation receives a request for tuples, it computes the next tuple (or tuples) to be returned, and then returns that tuple.
 - If the inputs of the operation are not pipelined, the next tuple(s) to be returned can be computed from the input relations, while the system keeps track of what has been returned so far.
 - If it has some pipelined inputs, the operation also makes requests for tuples from its pipelined inputs. Using the tuples received from its pipelined inputs, the operation computes tuples for its output, and passes them up to its parent.
 2. In a **producer-driven pipeline**, operations do not wait for requests to produce tuples, but instead generate the tuples **eagerly**.



- Each operation in a producer-driven pipeline is modeled as a separate process or thread within the system that takes a stream of tuples from its pipelined inputs and generates a stream of tuples for its output.

Data Security

- Data security is the protection of the data from unauthorized users.
- Only the authorized users are allowed to access the data.
- Most of the users are allowed to access a part of database i.e., the data that is related to them or related to their department.
- Mostly, the DBA or head of department can access all the data in the database.
- Some users may be permitted only to retrieve data, whereas others are allowed to retrieve as well as to update data.
- The database access is controlled by the DBA.
- He/she creates the accounts of users and gives rights to access the database.
- Users or group of users are given usernames protected by passwords.
- The user enters his/her account number (or user name) and password to access the data from database.
- For example, if you have an account in the "yahoo.com", then you have to give your correct username and password to access your account of e-mail.
- Similarly, when you insert your ATM card into the Automated Teller Machine (ATM), the machine reads your ID number printed on the card and then asks you to enter your pin code (or password). In this way you can access your account.

➤ What is the difference between security and integrity?

Data Security	Data Integrity
Data security defines a prevention of data corruption through the use of controlled access mechanism.	Data integrity defines a quality of data, which guarantees the data is complete and has a whole structure.
Data security deals with protection of data.	Data integrity deals with the validity of data.
Data security is making sure only the people who should have access to the data are the only ones who can access the data.	Data integrity refers to the structure of the data and how it matches the schema of the database.
Authentication/authorization, encryptions and masking are some of the popular means of data security.	Backing up, designing suitable user interface and error detection/correction in data are some of the means to preserve integrity.

➤ **What is authorization and authentication? OR What is difference between authorization and authentication?**

Authorization	Authentication
It is protecting the data to ensure privacy and access control data. Authorization is giving access to authorized users.	Authentication is providing integrity control and security to the data.
Authorization is the process of verifies what you are authorized to do or not to do.	Authentication is the process of verifying who you are.
Accessing a file from hard disk is authorization because the permissions are given to you to access that file allow you access it that is authorization.	Logging on to a PC with a username and password is authentication.

Audit Trail (audit log)

- An audit trail (audit log) is one record which will be generated against each and every transaction. Regarding the transaction, it will keep certain information.
- An audit trail (audit log) records;
 1. Who (user or the application program and a transaction number)
 2. When (date and time)
 3. From where (location of the user and/or terminal)
 4. What (identification of the data affected, as well as a before-and-after image of that portion of the database that was affected by the update operation)

Access Control

- Data security has to do with the protection of data against unauthorized access, while integrity has to do with data correctness. Security means protecting the data against unauthorized users and integrity means protecting the data against authorized users.
- In other words, security means making sure users are allowed to do the things they are trying to do; integrity means making sure the things they are trying to do are correct. Modern DBMSs typically support either or both of two broad approaches to data security, discretionary control and mandatory control. The two approaches differ in the following manner:
- In the case of discretionary control, a given user will typically have different access rights (privileges) on different objects; further, there are few inherent limitations regarding which users can have which rights on which objects (for example, user U1

might be able to see A but not B, while user U2 might be able to see B but not A). Discretionary schemes are thus very flexible.

- In the case of mandatory control, by contrast, each data object is labeled with a certain classification level, and each user is given a certain clearance level. A given data object can then be accessed only by users with the appropriate clearance. Mandatory schemes thus tend to be hierarchic in nature and hence comparatively rigid. (if U1 can see A but not B, then the classification of B must be higher than that of A, and so no user U2 can see B but not A).

⇒ **Authorization –Checking** a given access request against the applicable constraints in the catalog.

⇒ **Authentication –Process** of checking that the users are who they say they are.

Types of Access Control

1. Discretionary Access Control (DAC)

- File and Data Ownership
- DAC Access Modes:
 - ✓ **READ:** Allows user to read an object.
 - ✓ **WRITE-APPEND:** Allows user to expand object but not change previous content
 - ✓ **WRITE-CHANGE:** Allows user to modify and delete all or part of the contents of an object but does not allow a user to expand or view the object
 - ✓ **WRITE-UPDATE:** Allows user to modify the contents of an object but does not allow a user to add to, delete from, or view an object
 - ✓ **WRITE:** Allows user to modify, add, or delete the contents of an object in any manner but does not allow to view object
 - ✓ **EXECUTE:** Allows a subject to run the object as an executable file
 - ✓ **DELETE:** Allows to delete object
 - ✓ **NULL:** No access. Exclude a particular user
 - ✓ **Control:** Control objects, Control with passing ability

Access Control Matrix

Objects/User	Kimsfile	Donsfile	Payrol1	Payrol2	Doesfile
Kim	RW	R	RW	R	-
Joe	-	R	-	-	-
Don	-	RW	R	-	-
Jones	-	-	R	-	-
Doe					RW
Mgr	CP	CP	C	C	C
Jim			RW	RW	

W- Write, R-Read, CP- Control with passing ability

2. Mandatory Access Control (MAC)

- Important terms:

⇒ **Sensitivity Labels:** All subjects and objects must have labels assigned to them. A subject's sensitivity label specifies its level of trust. An object's sensitivity label specifies the level of trust required for access. In order to access a given object, the subject must have a sensitivity level equal to or higher than the requested object.

⇒ **Data Import and Export:** Controlling the import of information from other systems and export to other systems (including printers) is a critical function of MAC-based systems, which must ensure that sensitivity labels are properly maintained and implemented so that sensitive information is appropriately protected at all times.

➤ Difference between DAC and MAC

DAC	MAC
Allows users the ability to make policy decisions and/or assign security attributes.	Security policy is centrally controlled by a security policy administrator. Users do not have the ability to override the policy and, for example, grant access to files that would otherwise, to data entities.
Less secure than MAC	More secure than DAC. Used in military applications
Faster than MAC	Slower than DAC due to higher security

3. Role Based Access Control (RBAC)

- It is based on the concept that privileges and other permissions are associated with organizational roles, rather than individual users. Individual users are then assigned to appropriate roles.
- E.g., an accountant in a company will be assigned to the accountant role, gaining access to all the resources permitted for all accountants on the system. Similarly, a software engineer might be assigned to the Developer role.
- Roles are centrally managed by the administrator.
- Security groups are created representing each role and permissions and rights are assigned to these groups.
- Each user is added to a particular security group.
- RBAC is known as Non-Discretionary Access Control.

➤ Disadvantages

- It is necessary to understand each user's functionality in depth so that roles can be properly assigned.
- If roles are not assigned properly then inappropriate access right creates security problems.

➤ Advantages

- The security is easily maintained by limiting access to sensitive information based on the security groups.



- Roles can be aligned with the organizational structure of the business.

4. Rule-based access control (RuBAC)

- With the rule-based model, a security professional or system administrator sets access management rules that can allow or deny user access to specific areas, regardless of an employee's other permissions.